

Ultrasonic Distance Collection

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1. Experiment Purpose

The Lastbit car is equipped with an ultrasonic sensor in the front. This lesson will guide you through learning how to read data from the ultrasonic sensor.

The ultrasonic sensor is a sensor developed based on the characteristics of ultrasonic waves. When operating, the `TRIG` pin emits an ultrasonic signal, which bounces back when it encounters an obstacle. When the `ECHO` pin receives the reflected ultrasonic signal, it transmits the signal to the controller for processing. The controller calculates the time required from transmission to reception to determine the distance between the probe and the object. The audible sound wave frequency for the human ear ranges from 20Hz to 20KHz. The ultrasonic module emits sound at a frequency greater than 20KHz, so the human ear cannot hear the sound from the ultrasonic module.



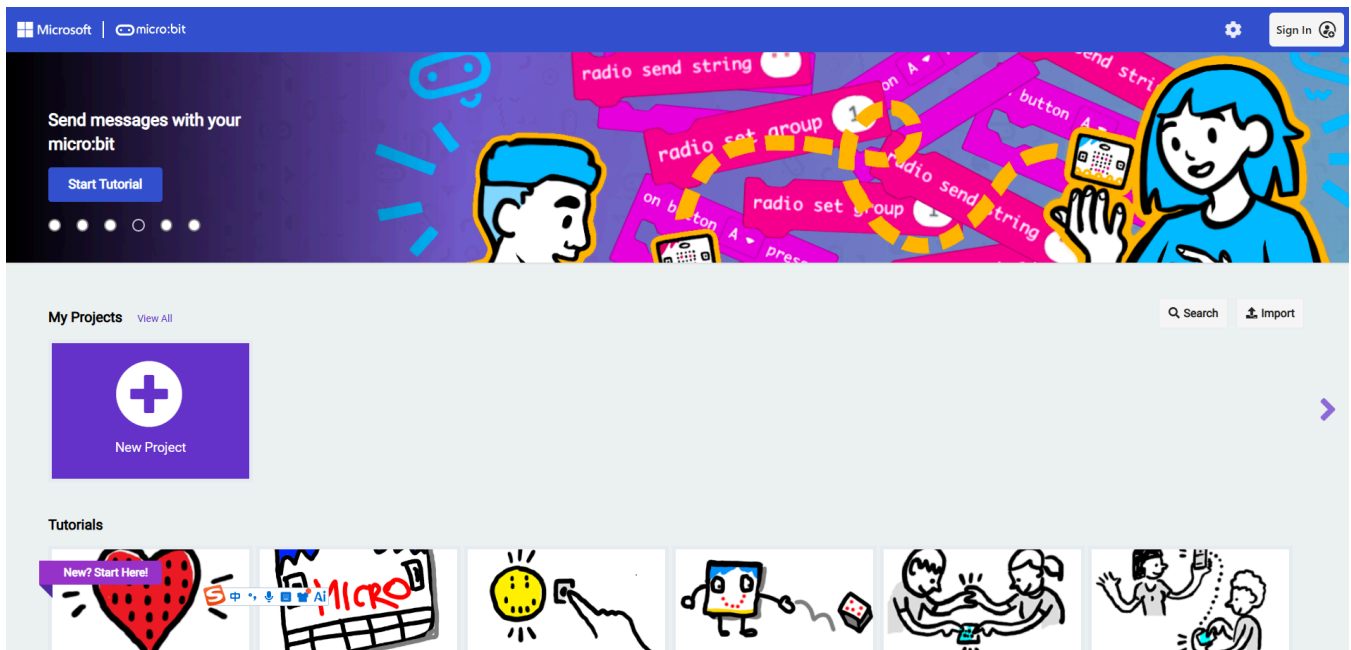
2. Experiment Steps

1. Before starting the experiment, connect the Micro:bit mainboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.



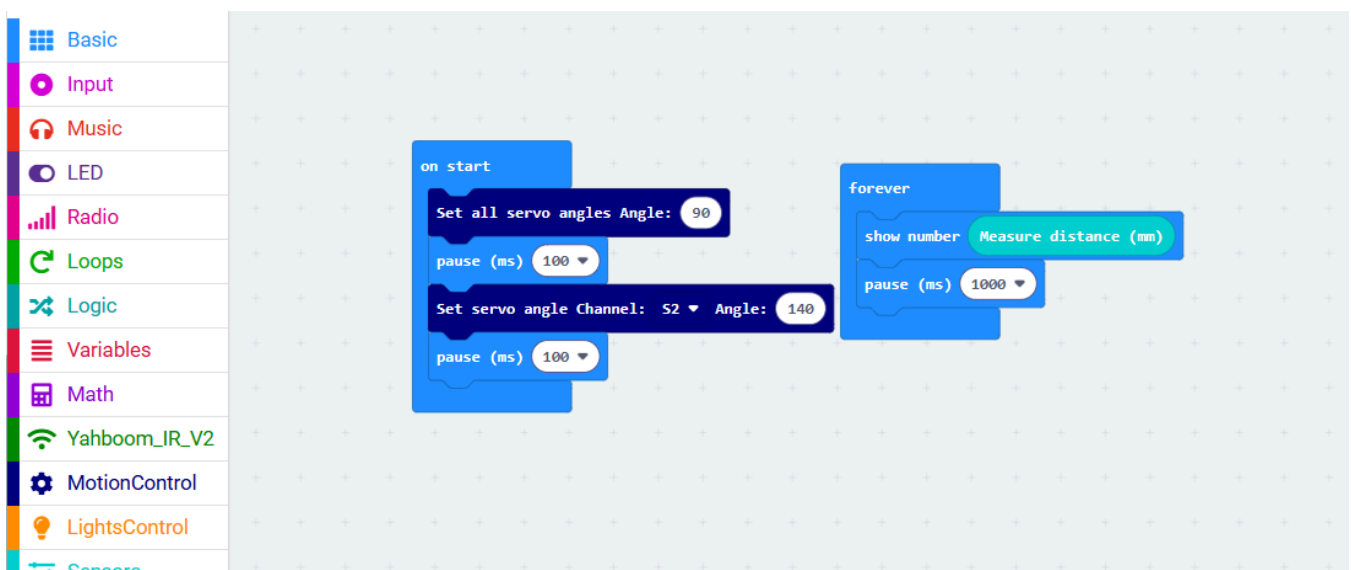
2. Open MakeCode using the link below.

[Microsoft MakeCode for micro:bit](#)

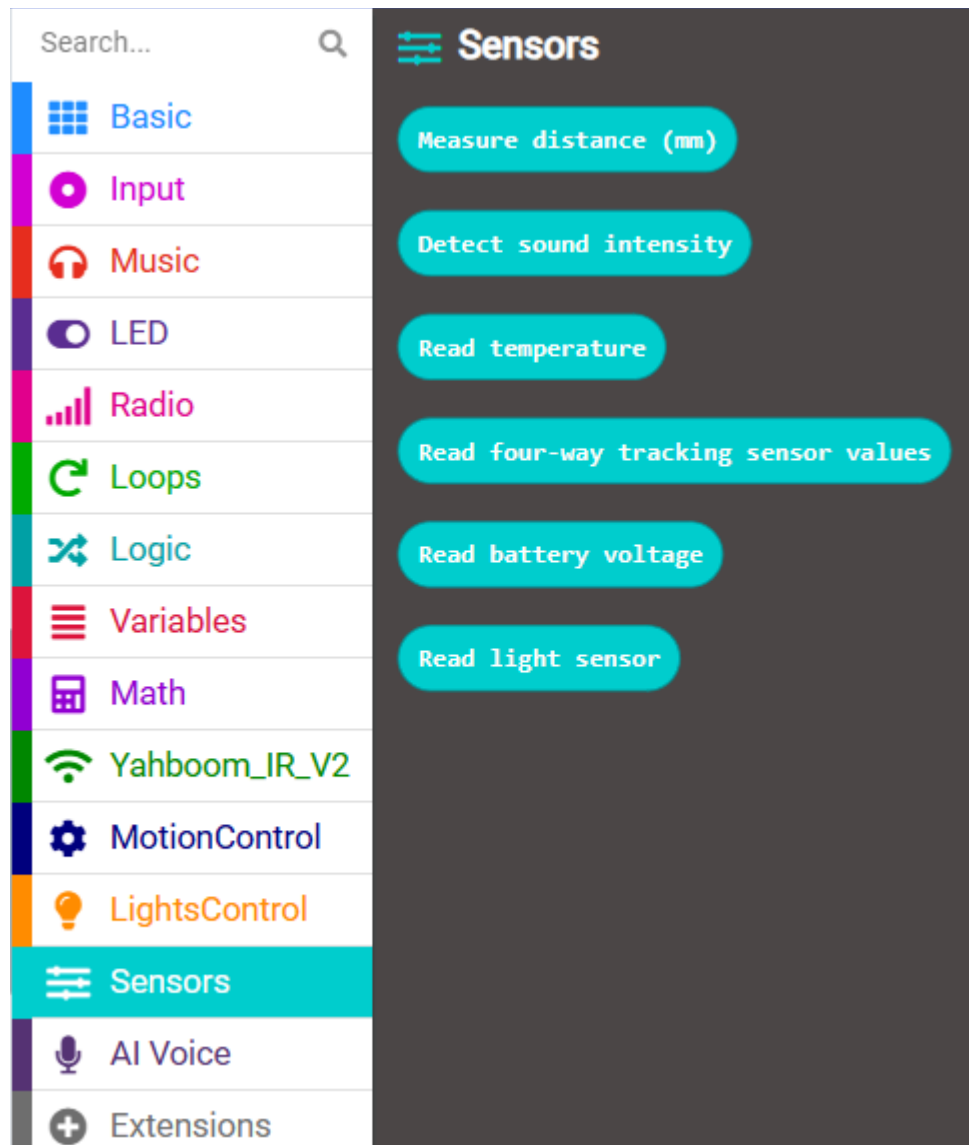


3. Drag `1.Ultrasonic_Distance_Collection.hex` to the new project page. MakeCode will automatically import and open the project.

3. Experiment Source Code

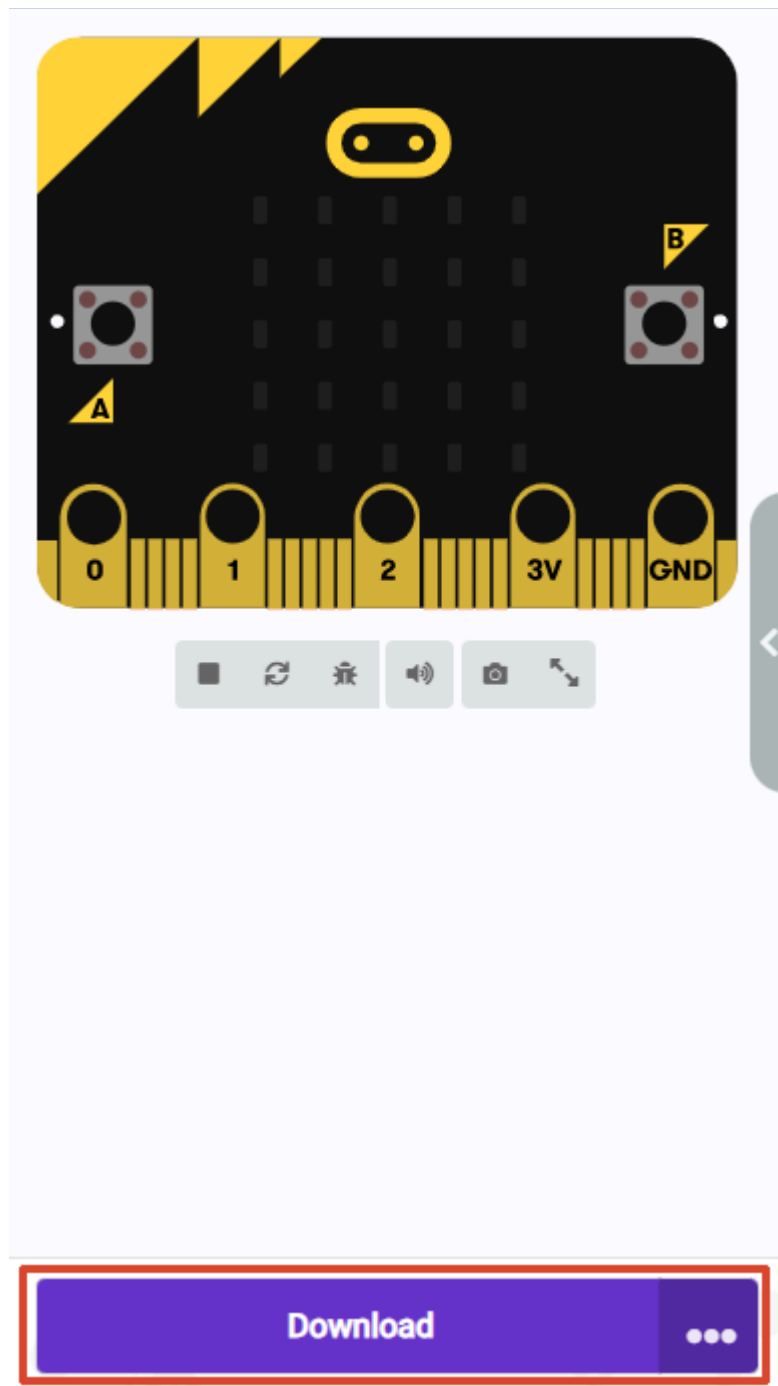


This lesson's building blocks are located in the `sensors` category.



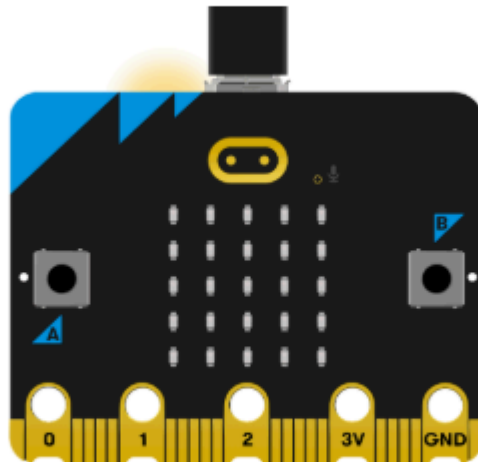
4. Program Download

1. Click the download button at the bottom left to download the program to the Micro:bit.



2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please connect the data cable to the Microbit first. If already connected, click [Next] here.

1. Connect your micro:bit to your computer

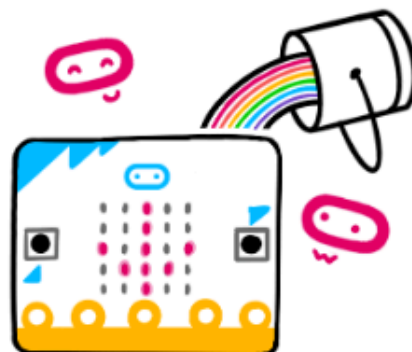


 Next

3. If the Microbit is detected as properly connected, the following interface will be displayed. At this point, you can click [Download].

 **Connected to micro:bit** 

Your micro:bit is connected! Pressing 'Download' will now automatically copy your code to your micro:bit.



 Download

If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. You can choose between two methods: [Download as File] and [Pair].

2. Pair your micro:bit to your browser



Press the Pair button below.

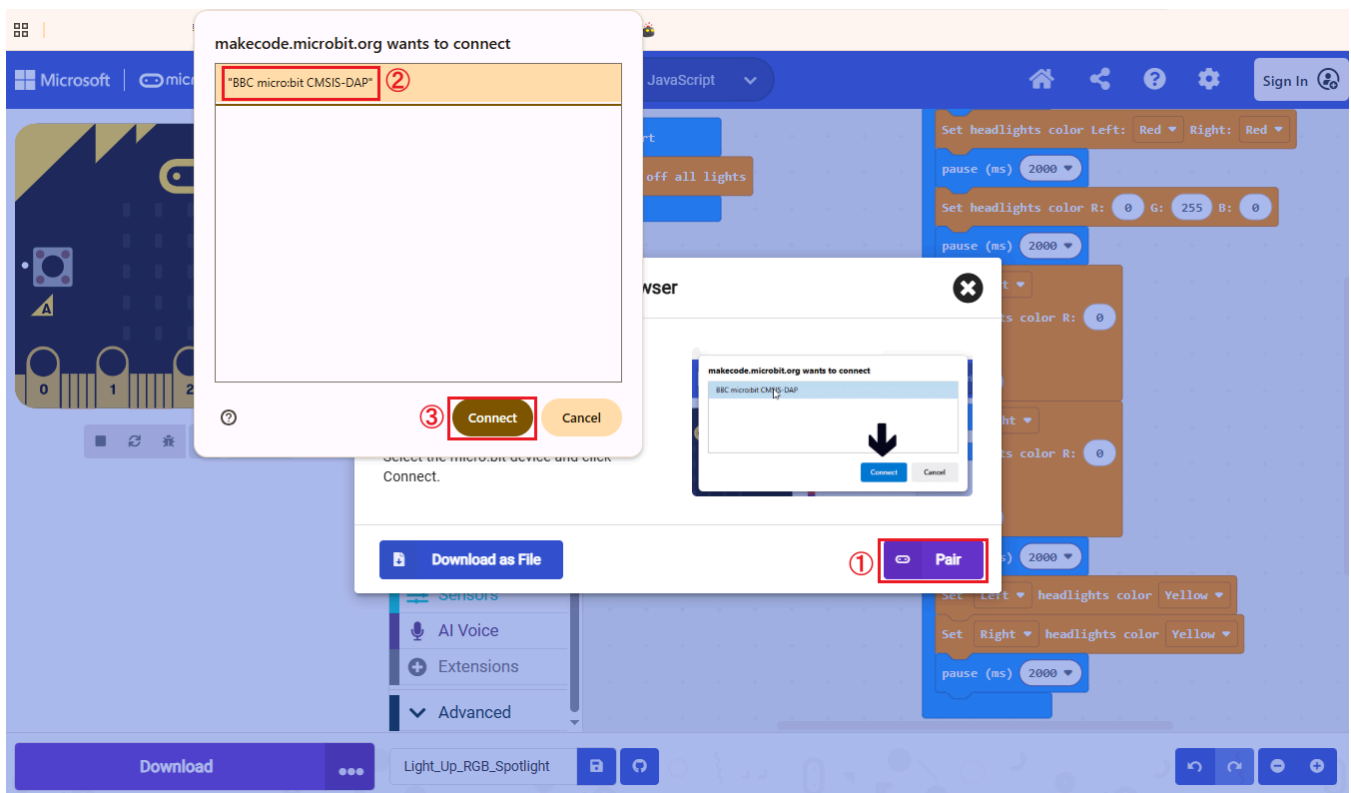
A window will appear in the top of your browser.

Select the micro:bit device and click Connect.

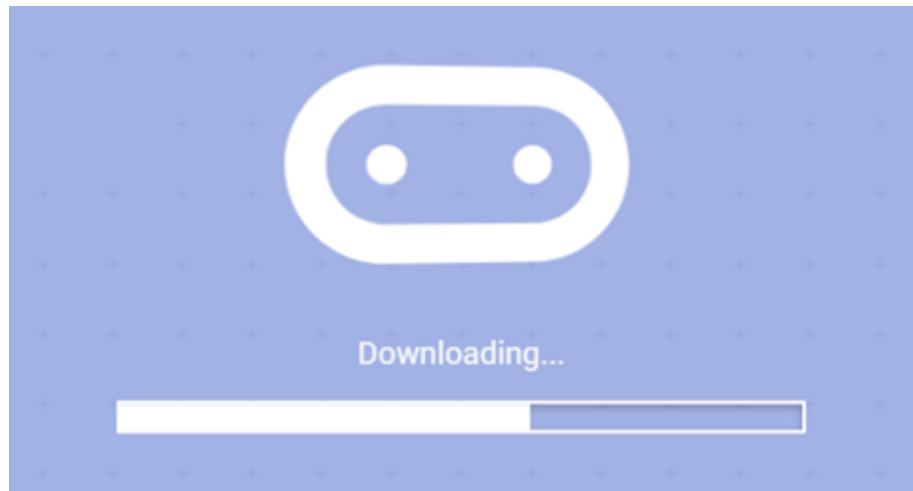


When selecting [Download as File], the system will download the currently opened program as `microbit-project-name.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive letter to complete the download.

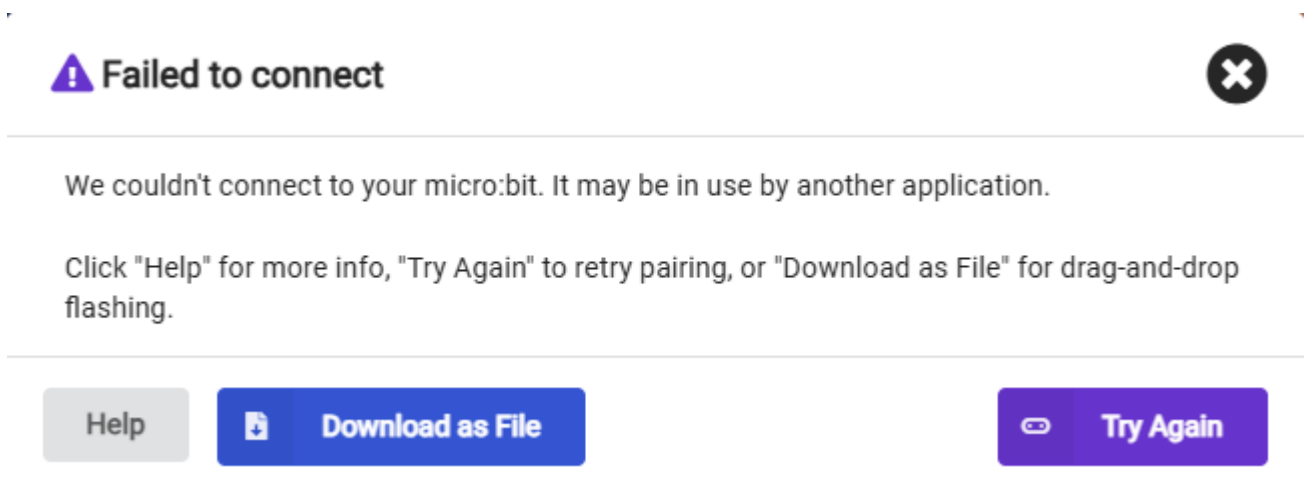
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC micro:bit CMSIS:DAP`.



4. When the following interface appears, it means the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



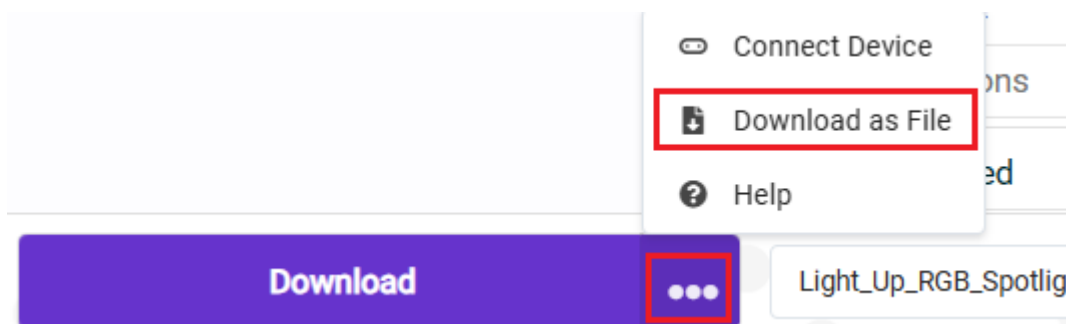
If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.



If you want to save the file directly, we provide two methods:

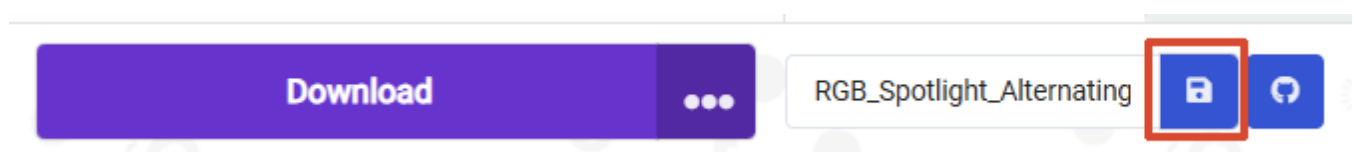
Method 1:

Click the [...] next to download, then select [Download as File]



Method 2:

Click [Save] after the [Project Name Edit Field].



5. Experiment Results

After flashing, turn on the switch (flip the switch to the ON position). The Micro:bit LED matrix screen will display the distance measured by the ultrasonic sensor in millimeters (mm). If you need the unit in centimeters, you can perform a unit conversion.